QUESTION LESSON 4 AND 5 ING

1. Concept of scalability and concept of interoperability in blockchain world

2. Sharding in Blockchain

Explain how sharding works as a method to enhance blockchain performance. What are the main technical and security challenges that sharding introduces?

3. Decentralized Consensus and Information Asymmetry

Decentralized consensus mechanisms aim to eliminate the need for trusted intermediaries, yet information asymmetry can still exist among participants. Discuss how consensus protocols address (or fail to address) the problem of information asymmetry in distributed networks.

4. Proof of Work (PoW) vs. Proof of Stake (PoS)

Compare and contrast Proof of Work and Proof of Stake as consensus mechanisms.

OPEN QUESTIONS LESSON 6 AND 7

1. How does the problem of **asymmetric information** influence the process of **distributed consensus** in blockchain networks, and in what ways does blockchain technology attempt to reduce or eliminate this imbalance?

2. What is the role of **innovators and managers** in promoting and implementing **smart contracts** within an organization, and what challenges might they face in aligning technological innovation with organizational goals and governance structures?

3. In what ways could the adoption of **large, standardized smart contract templates** facilitate **mass adoption and efficiency** in industries that rely on frequent and large-scale transactions? What risks might this standardization introduce?

4. Compare the **Proof of Work (PoW)** and **Proof of Stake (PoS)** mechanisms in terms of **security, energy efficiency, and decentralization**. What are the main benefits and drawbacks of each, and how do they affect the sustainability and scalability of blockchain systems?